Synthetic Health Data Evaluation Report

Generated on: March 16, 2026

Datasets Used in Evaluation

Type	Filename
Dataset 1 (Real)	original_preprocessed.csv
Dataset 2 (Synthetic)	synthetic_cgan_preprocessed.csv

Predictive Models Used

Model File
logistic_regression_original.pkl

1. DPCM2 Evaluation

DPCM2 Overall Score: 91.07%

Feature	Score
age	0.986
bmi	0.984
systolic_bp	0.872
diastolic_bp	0.755
family_diabetes	0.860
family_hypertension	0.836

Higher DPCM2 scores indicate stronger similarity between original and synthetic datasets.

2. Resemblance Analysis

Test	Result
Jensen-Shannon Divergence	0.2852
Kolmogorov-Smirnov D-Statistic	0.1230

Lower values indicate closer similarity between distributions.

3. Utility Evaluation

TSTR (Train Synthetic, Test Real)

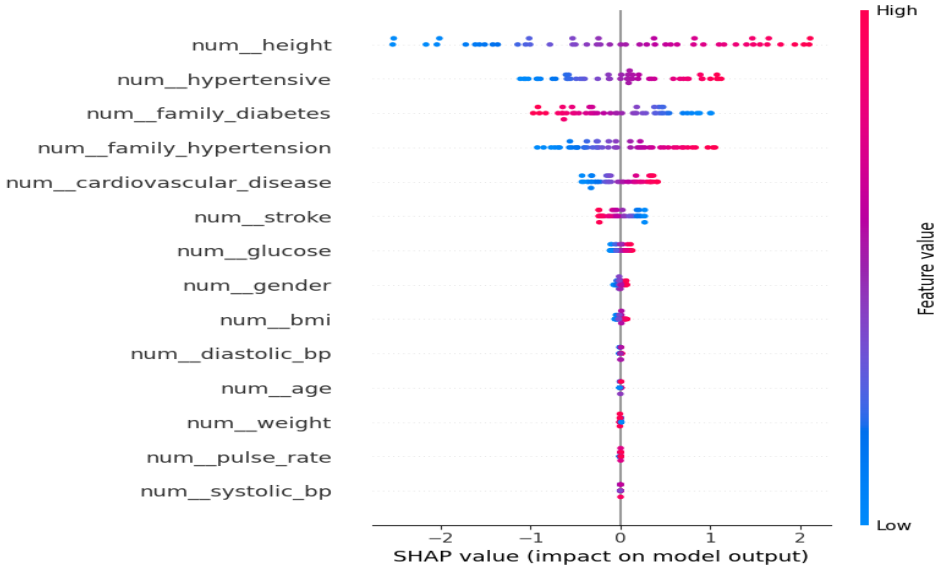
Metric	Score
Accuracy	0.903
AUC	0.789
Recall	0.327
Precision	0.282
F1-Score	0.303

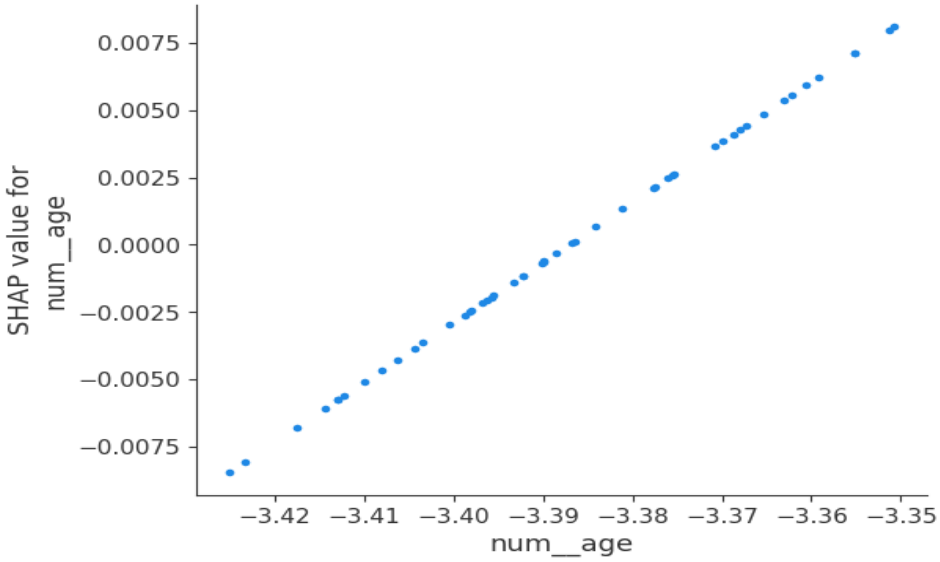
TRTR (Train Real, Test Real)

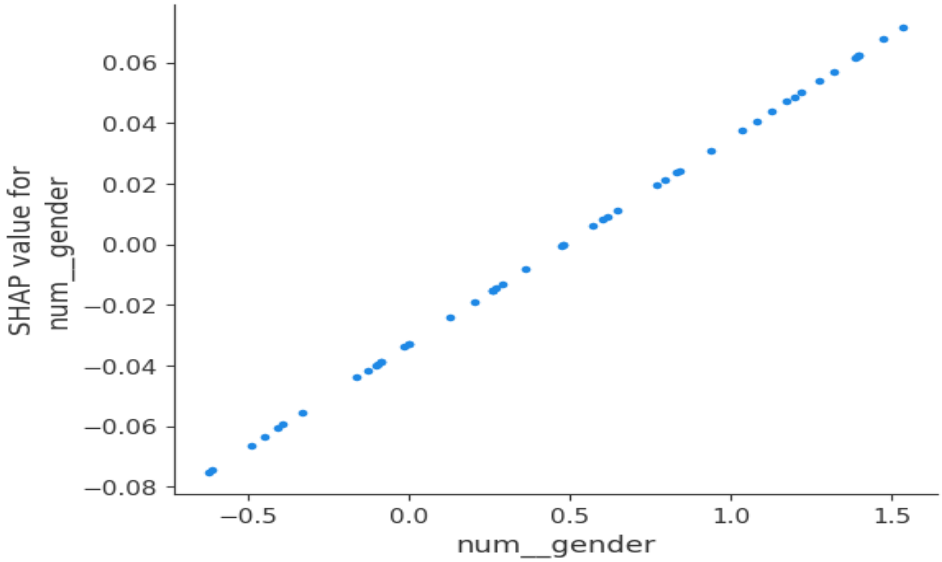
Metric	Score
Accuracy	0.939
AUC	0.858
Recall	0.263
Precision	0.756
F1-Score	0.390

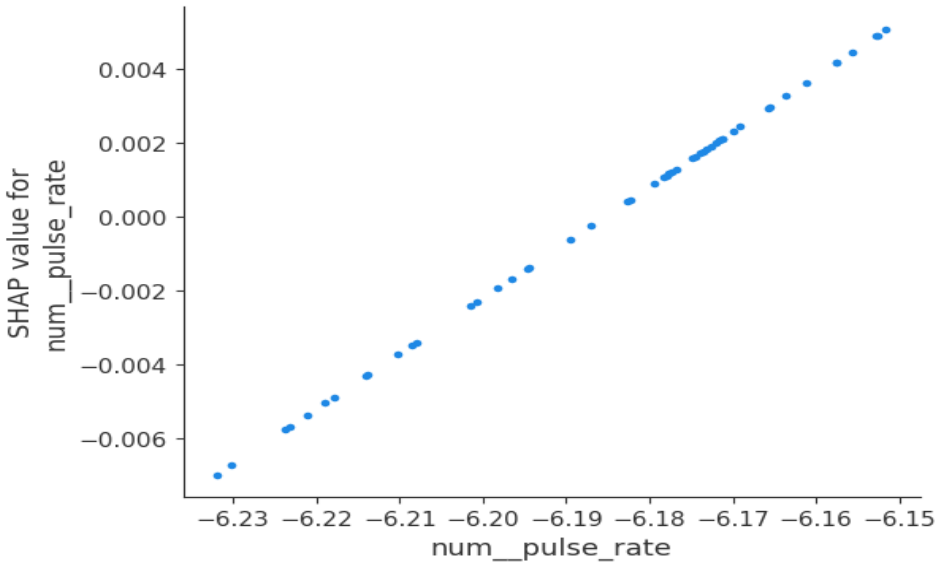
TSTR measures whether synthetic data can train models that generalize to real-world cases. TRTR provides the real-data baseline.

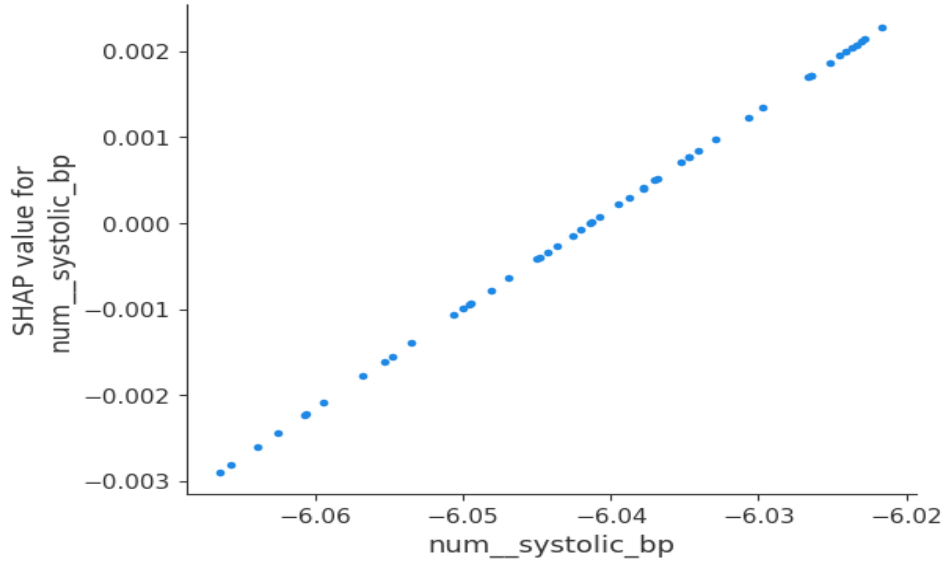
4. Explainable AI (SHAP)

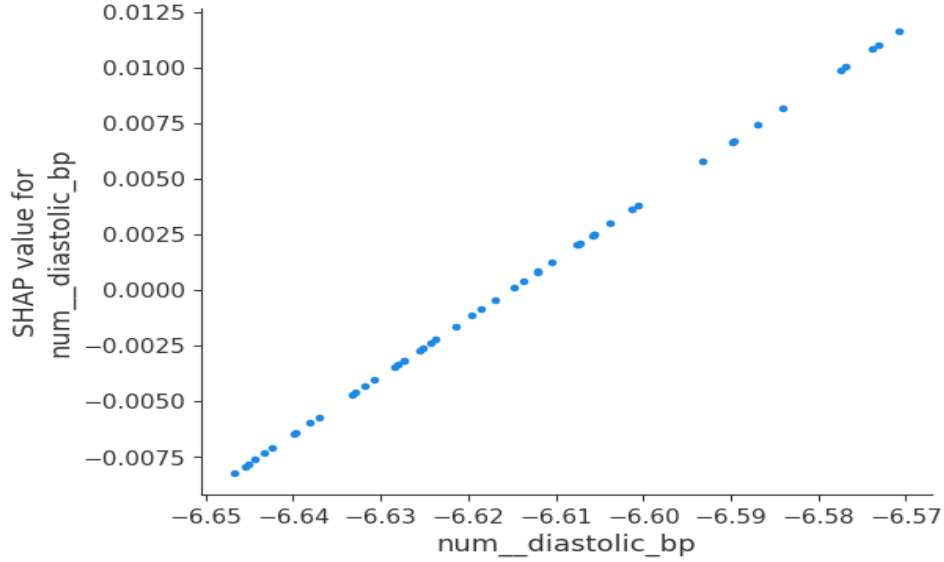

num__height
num__hypertensive
num__family_diabetes
num__family_hypertension
num__cardiovascular_disease
num__stroke
num__glucose
num__gender
num__bmi
num__diastolic_bp
num__age
num__weight
num__pulse_rate
num__systolic_bp
−2
−1
0
1
2
SHAP value (impact on model output)
High
Feature value
Low

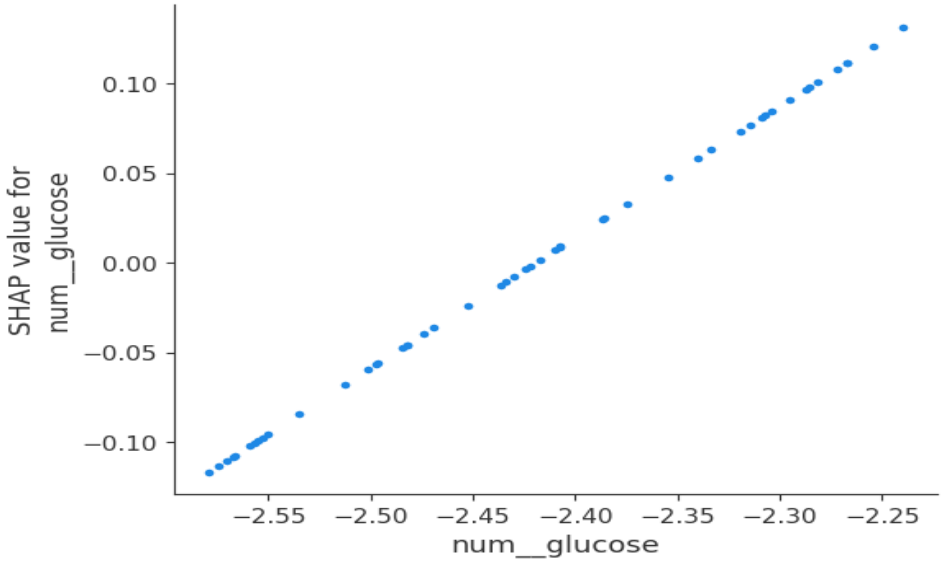

0.0075
0.0050
0.0025
0.0000
−0.0025
−0.0050
−0.0075
SHAP value for num__age
−3.42
−3.41
−3.40
−3.39
−3.38
−3.37
−3.36
−3.35
num__age

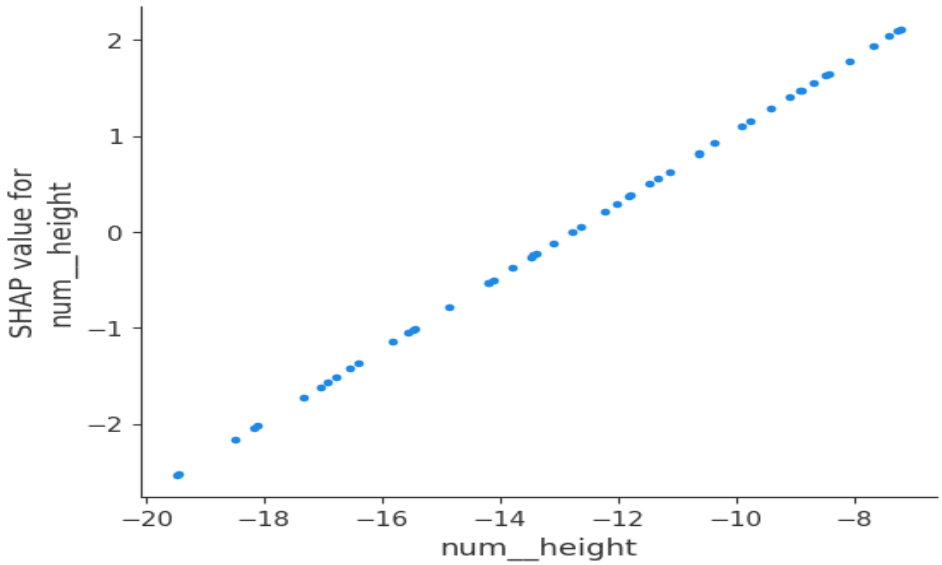

SHAP value for
num__gender
0.06
0.04
0.02
0.00
−0.02
−0.04
−0.06
−0.08
−0.5
0.0
0.5
1.0
1.5
num__gender

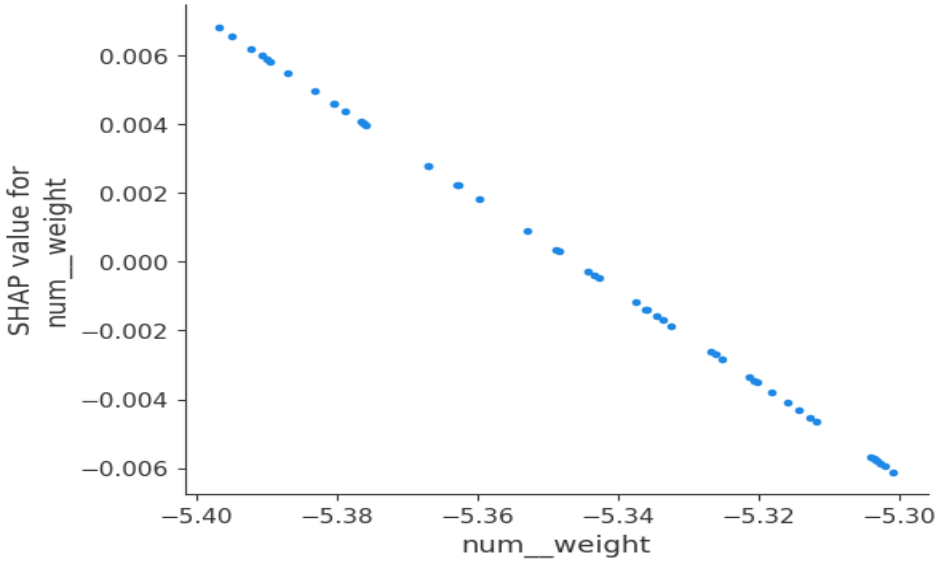

SHAP value for
num__pulse_rate
0.004
0.002
0.000
−0.002
−0.004
−0.006
−6.23
−6.22
−6.21
−6.20
−6.19
−6.18
−6.17
−6.16
−6.15
num__pulse_rate

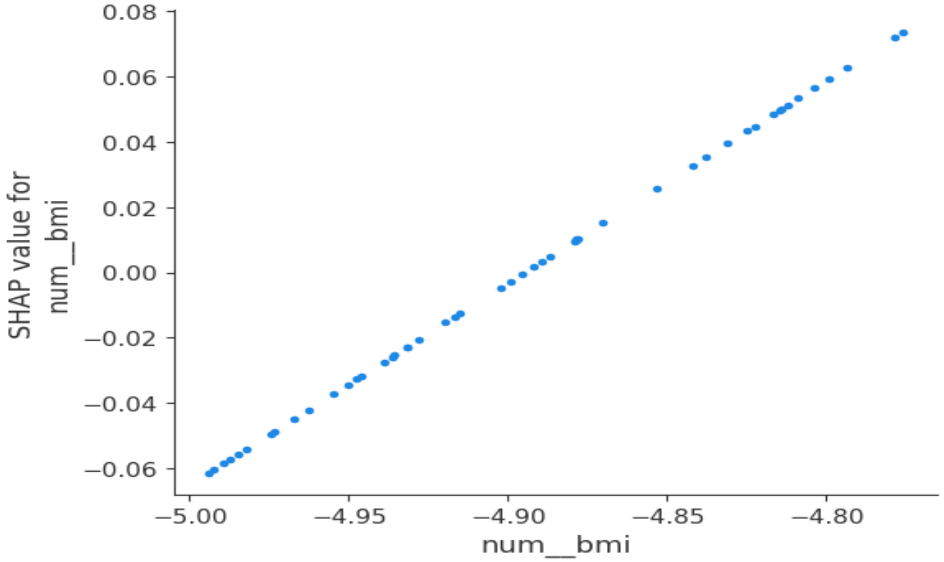

SHAP value for
num__systolic_bp
0.002
0.001
0.000
−0.001
−0.002
−0.003
−6.06
−6.05
−6.04
−6.03
−6.02
num__systolic_bp

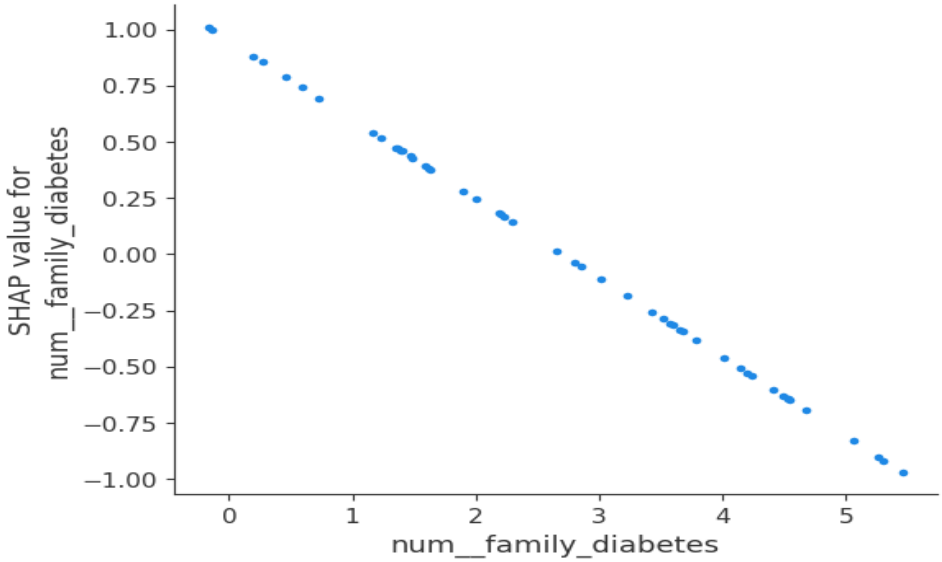

SHAP value for
num__diastolic_bp
0.0125
0.0100
0.0075
0.0050
0.0025
0.0000
−0.0025
−0.0050
−0.0075
−6.65
−6.64
−6.63
−6.62
−6.61
−6.60
−6.59
−6.58
−6.57
num__diastolic_bp

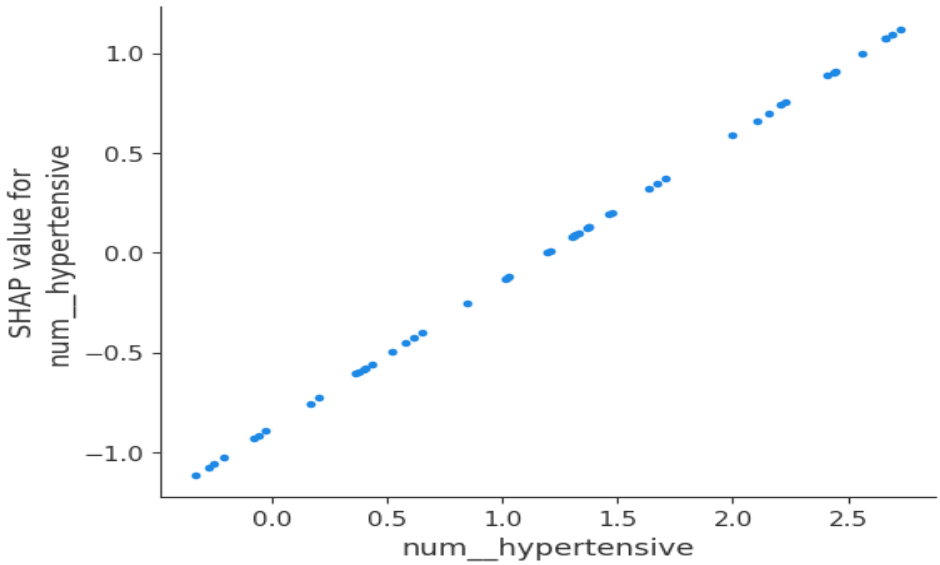

SHAP value for
num__glucose
0.10
0.05
0.00
−0.05
−0.10
−2.55
−2.50
−2.45
−2.40
−2.35
−2.30
−2.25
num__glucose

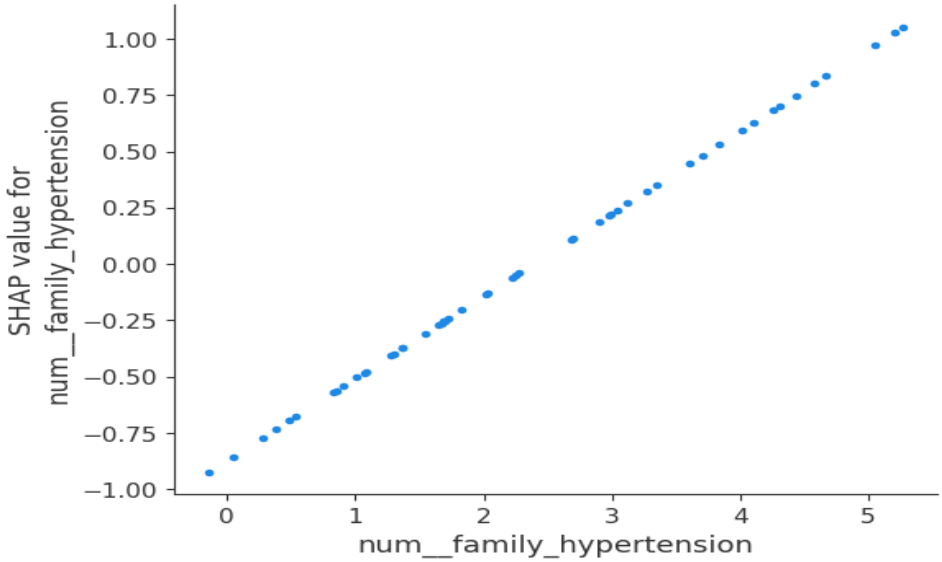

SHAP value for
num__height
2
1
0
−1
−2
−20
−18
−16
−14
−12
−10
−8
num__height

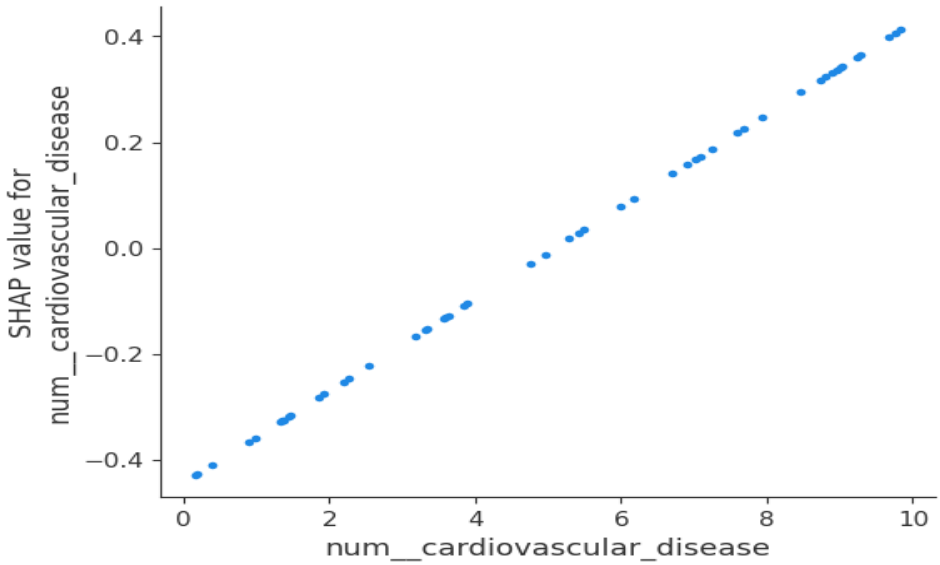

SHAP value for
num__weight
0.006
0.004
0.002
0.000
−0.002
−0.004
−0.006
−5.40
−5.38
−5.36
−5.34
−5.32
−5.30
num__weight

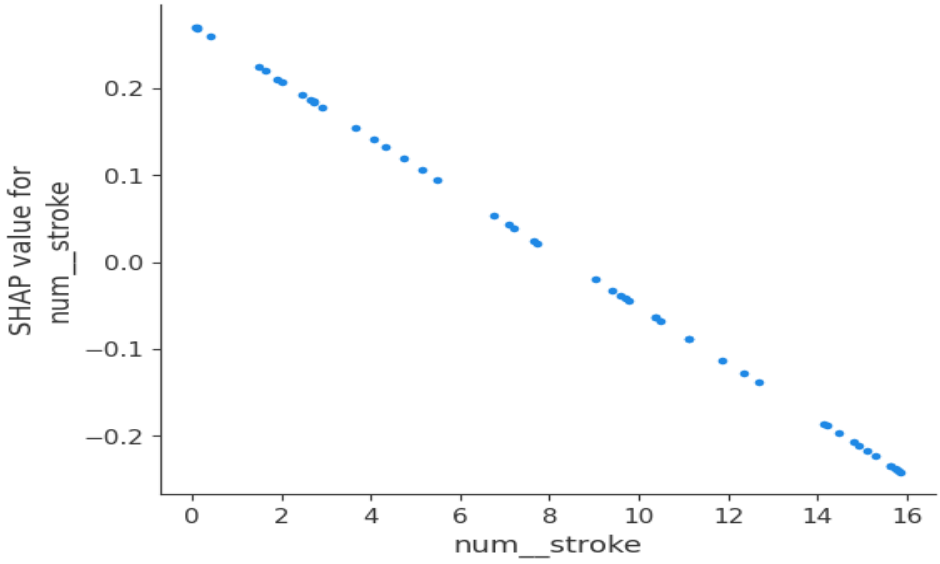

SHAP value for
num__bmi
0.08
0.06
0.04
0.02
0.00
−0.02
−0.04
−0.06
−5.00
−4.95
−4.90
−4.85
−4.80
num__bmi


SHAP value for
num__family_diabetes
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00
0
1
2
3
4
5
num__family_diabetes


SHAP value for
num__hypertensive
1.0
0.5
0.0
−0.5
−1.0
0.0
0.5
1.0
1.5
2.0
2.5
num__hypertensive


SHAP value for
num__family_hypertension
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00
0
1
2
3
4
5
num__family_hypertension


SHAP value for
num__cardiovascular_disease
0.4
0.2
0.0
−0.2
−0.4
0
2
4
6
8
10
num__cardiovascular_disease



SHAP plots show feature influence on model predictions.